CHAPTER

Artificial intelligence in animal farms for management and breeding

11

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Introduction

Livestock farming plays a significant role in meeting the global demand for food, supporting livelihoods, and contributing to economic development (Banda and Tanganyika, 2021; Hamadani et al., 2022a). Livestock provides a valuable source of protein, including meat, poultry, fish, eggs, and dairy products, which are crucial components of a balanced diet. Livestock farming helps promote food security and provides nutritional diversity to meet the dietary needs of a growing global population (Reynolds et al., 2015; Varijakshapanicker et al., 2019). It also contributes to the cycling of nutrients in agricultural systems. Livestock waste, such as manure, can be used as organic fertilizer (vermicompost), replenishing the soil with essential nutrients and improving soil fertility. This nutrient cycling helps maintain soil health, increases agricultural productivity, reduces the need for synthetic fertilizers, and contributes to the cycling of nutrients in agricultural systems. Additionally, vermicompost production also adds to the farmer's income (Hamadani et al., 2020d).

Livestock can utilize land resources that are not suitable for crop cultivation. Grazing animals on pasturelands and utilizing marginal lands for forage production can efficiently convert nonarable land into valuable food resources. They are capable of converting low-quality protein into high-quality protein (Broderick, 2018). Integrating animal farming with crop production through mixed farming systems can promote sustainable land use and optimize resource utilization (Hamadani et al., 2021b). Various by-products from the agriculture and food industries are utilized in livestock farming, which helps in reducing waste, promoting sustainability, and socioeconomic upliftment (Hamadani et al., 2020c; Hamadani, 2023). For example, animal feed can be produced from agricultural residues, food processing by-products (Bhandari and Bahadur, 2019), and other biomass that would otherwise go to waste. This minimizes environmental impact and improves resource efficiency.

Livestock farming contributes to national and international trade, creating economic opportunities and generating foreign exchange earnings (Baltenweck et al., 2020). Countries with efficient livestock sectors can export meat, dairy products, and other animal-derived commodities, contributing to their overall economic growth and trade balance. Conventional and alternate livestock and poultry resources

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contribute to this. The cultural and social significance of animal farming can be also adjudged by the fact that it is rooted in the cultural and social fabric of many societies. Livestock and poultry rearing is often intertwined with traditional practices, festivals, and cultural celebrations by playing a role in preserving cultural heritage, supporting rural communities, and maintaining social cohesion (Alders et al., 2021; Hamadani et al., 2020e; Hamadani and Khan, 2013).

Despite all this, livestock farming today faces challenges related to sustainability, animal welfare, and environmental impact (Godde et al., 2020), which were irrelevant a few decades back. These include population explosion, climate change, antibiotic resistance, new and emerging animal diseases, food insecurity, loss of genetic variance, depleting and degrading resources, etc (Eeswaran et al., 2022). It is important to address these challenges through responsible and sustainable practices, such as implementing efficient resource management, adopting animal welfare standards, and embracing innovative technologies that minimize environmental footprint. Additionally, some major constraints faced in livestock management and breeding is that it is labor intensive and time-consuming and involves drudgery of repetitive tasks leading to inefficiency and low productivity.

Artificial intelligence (AI) has the potential to revolutionize livestock management and breeding with its vast applicability. The algorithms of machine learning (ML) are helping to transform the conventional inefficient system of farming into smart farming. Livestock is monitored and managed smartly with AI-based systems to enhance animal welfare, improve health, control diseases, increase production, and provide useful insights to farm managers.

AI and big data in livestock farms

Artificial intelligence has the potential to significantly improve all aspects of animal farming practices by providing innovative solutions for monitoring, management, and decision-making (Hamadani et al., 2023). However, before the major applications of AI in animal management and breeding are discussed, it is important to understand the relevance of big data in animal management (Hamadani et al., 2020b). Big data in animal farms empowers farmers and offers valuable insights and tools to improve animal health, productivity, and overall farm management. By leveraging the power of data analytics, farmers can make informed decisions, enhance efficiency, and contribute to sustainable and profitable animal farming practices. AI is the obvious choice for the analysis of data generated because it deals very efficiently with data that has volume, variety, and velocity (Wong et al., 2019). Under the conditions of large volumes of data, AI allows the delegation of difficult pattern recognition, learning, and other tasks to computer-based approaches. In addition, AI can handle the great velocity of data, by enabling rapid computer-based decisions each leading to other subsequent decisions. Variety is mitigated by capturing, structuring, and understanding unstructured data using AI and other analytics.

Artificial intelligence consists of all computing systems that can perform tasks that require human intelligence, such as decision-making, object detection, solving complex problems, and so on. The realm of AI also prompts the inception of ML and deep learning, as well as predictive analytics (Wong et al., 2019). All this is collectively referred to as data science. This is a data-intensive science, and the data gathered do not only benefit the farm but also researchers, companies, and other stakeholders use it to make inferences that can help farms around the world to make informed decisions on individual animal performance.

Artificial intelligence includes various techniques viz. ML, deep learning, natural language processing, fuzzy logic, robotics, and expert systems. ML is a subcategory of AI, which is mainly divided into supervised learning, reinforcement learning, and unsupervised learning (Rashidi et al., 2019). Supervised learning is a very popular area of ML in animal sciences. Artificial neural networks also come under the umbrella of ML, which is used both for supervised and unsupervised learning. These derive inspiration from the natural neural network of the human nervous system. Deep learning is the process of the implementation of neural networks on high-dimensional data to gain insights and form solutions. Data from livestock are currently being used for satellite imagery, data tracking, and analysis applications (Hamadani and Ganai, 2022). It is proving to be useful for the improvement of animal health and food safety, identification, traceability, and labor reduction. The most prominent areas where AI is being used to bring about improvement are discussed below.

Identification of animals

Animal identification involves identifying animals individually, which allows monitoring and tracking each animal separately. It has a variety of applications, which include record keeping, efficient farm management, ownership verification, biosecurity control, registration, insurance, and security of animals from theft. Identification of livestock is also needed by different regulatory bodies, which ensure safety and traceability and help in improving the product quality (Fuentes et al., 2022; Hossain et al., 2022). In many livestock farms, the path of each animal is monitored from birth onward by using an identification system that helps in keeping track of different parameters and maintains the information base of each animal on the farm.

In earlier times, small-scale farmers would remember few important details about their animals without any proper record system. However, in some organized farms, manual record system would be maintained to keep track of the animals and which would also help in decision-making. Now with the modernization of farming practices and increased use of automated technology, the use of electronic identification systems is becoming necessary, especially in large commercial livestock farms. Many management operations are automatized in precision and smart farming leading to increased efficiency and reduced labor.

Automation on livestock farms tailors activities to the needs of individual animals, which need the subsystems to recognize the animals individually as they interact with these systems (Hamadani and Khan, 2015). Hence, electronic identification of animals becomes a prerequisite for all these systems to generate data on each animal and record that information for further processing to draw useful inferences. Integration of AI into these systems helps track animals intelligently with increased ease and efficiency. Identification of animals can be automatized by engaging computer vision. All the required information about individual animals can be accessed easily. Even the smaller species like poultry can also be identified individually using AI. AI-driven software is now being developed, which has the potential to noninvasively and unobtrusively track individual animals' location as well as estimate their position in an area (Congdon et al., 2022).

In addition to identification, AI has also been used in counting as well as in describing animals using camera-trap images (Norouzzadeh et al., 2018). Applications of AI on the data obtained by a thermal camera for locating animals during nocturnal hours have also been reported (Munian et al., 2022). This can be helpful in locating predators on livestock farms and preventing the loss of livestock

due to predation. Detection of cattle using images captured from drones has also been attempted (Rivas et al., 2018). Acoustics has been used to identify gender (Hamadani et al., 2017), breed, and species of animals and birds, and the same can be automatized by employing the techniques of AI (Yeo et al., 2011; Pabico et al., 2015).

Animal monitoring

Monitoring animal behavior and physiology is important for their welfare and management. Any abnormality detected in the behavior or physiology indicates either disease or any other abnormal situation which needs attention. Animals when managed in a way that their welfare is taken into consideration perform better. However, it is not easy to continuously monitor animal behavior and physiology manually. AI can be employed to continuously monitor animal behavior, physiological parameters, and environmental conditions (Congdon et al., 2022). Computer vision and AI have been employed for automatic behavior analysis in animals (Bernardes et al., 2021). ML algorithms can analyze large datasets generated by sensors, wearable devices, and video surveillance to detect early signs of illness, stress, or abnormal behavior. This allows for timely intervention and preventive measures to ensure optimal animal health (Ezanno et al., 2021). AI algorithms can analyze this data in real time to detect abnormalities, stress, or signs of illness. Farmers can receive timely alerts and take appropriate actions to ensure the health and well-being of their animals. This would be useful for AI can leverage historical and real-time data to develop predictive models for livestock health and performance. By analyzing factors such as genetics, environmental conditions, feed quality, and management practices, AI can generate insights and forecasts regarding animal growth, production outcomes, and disease susceptibility. This enables farmers to make informed decisions on breeding, nutrition, and overall management strategies. Various sensors embedded in wearable devices can collect data on animal behavior, health, and activity. Various sensors used in animal monitoring are given in Fig. 11.1. For example, sensors like image analysis sensors, walk-over-weigh sensors, GPS, acetometers, and thermometers are used for live weight estimation; calculation of reproductive efficiency, growth rates and maternal parentage; monitoring of calving, activity, health, rumination, lameness, body temperature, intake, heat detection, resting, feeding, calving, etc (Aquilani et al., 2022). While it remains relatively easy to implement in barn and pasture settings, free-range farming presents additional challenges; herd supervision can become more difficult as cattle are not contained. Supervision of animals in the barn or shed is relatively easy, while it is difficult in the pasture setup. To address this issue, virtual fence collars are used to remotely monitor and manage livestock in grazing areas (Versluijs et al., 2023). A sensor suite deployed with long-range radio (LoRa) can be used for tracking the activity and location of livestock in pasture lands (Dos Reis et al., 2021).

Disease detection and prevention

Artificial intelligence algorithms can analyze complex data patterns, including environmental factors, animal health records, and disease outbreaks, to predict and detect disease risks. By utilizing big data analytics and ML, AI can provide early warnings of potential disease outbreaks, enabling farmers to take proactive measures such as quarantine, vaccination, or treatment, thus reducing the spread of diseases and minimizing economic losses. These can help in the following:



FIGURE 11.1

Various sensors for animal monitoring.

- **Early disease detection:** AI algorithms can analyze various data sources, including animal health records, environmental factors, and sensor data, to detect early signs of disease. By identifying patterns, anomalies, and correlations in the data, AI can provide early warnings of potential disease outbreaks. This allows farmers to take prompt action, such as isolating affected animals, implementing biosecurity measures, or initiating targeted treatment protocols.
- **Real-time monitoring:** AI-powered systems can continuously monitor animal health using sensors, cameras, and wearable devices. These technologies can capture data on vital signs, behavior patterns, feed intake, and other health-related indicators. AI algorithms can process these data in real time, detecting deviations from normal behavior or physiological ranges. Real-time monitoring enables immediate response to any health issues, minimizing the spread of diseases and reducing the impact on overall herd or flock health.

- **Disease risk assessment:** AI can assist in assessing disease risks by analyzing a combination of historical data, environmental factors, and disease prevalence information. By considering variables such as weather conditions, geographical factors, animal movements, and biosecurity practices, AI algorithms can identify high-risk areas or periods for specific diseases. This information helps farmers prioritize prevention measures and allocate resources accordingly.
- **Disease prediction and modeling:** AI can utilize ML techniques to develop predictive models for disease outbreaks. By analyzing historical data, such as disease incidence, environmental conditions, and animal population dynamics, AI algorithms can forecast the likelihood and severity of future disease outbreaks. Such models enable farmers to take proactive measures, such as adjusting management practices or implementing preventive measures, to minimize the impact of diseases on animal health and productivity.
- **Surveillance and early warning systems:** AI can contribute to surveillance systems for animal diseases by analyzing diverse data sources, including animal health databases, laboratory reports, and social media data. By employing natural language processing and ML algorithms, AI can detect signals or outbreaks of emerging or contagious diseases. Early warning systems powered by AI can alert authorities and stakeholders, facilitating rapid response and containment of diseases.

Precision nutrition and feed management

AI can optimize animal nutrition by analyzing data on animal characteristics, feed ingredients, and environmental factors. By considering factors such as animal genetics, growth stage, and performance goals, AI algorithms can recommend personalized nutrition plans and feed formulations. This helps maximize feed efficiency, improve animal growth rates, and reduce waste, resulting in more sustainable, cost-effective feeding practices for personalized animal nutrition (Saxena and Parasher, 2019).

Artificial intelligence can significantly enhance precision nutrition and feed management in animal farming. Broadly, the application of AI is stated as under:

- **Nutrient analysis and formulation:** AI algorithms can analyze vast amounts of data, including animal characteristics, dietary requirements, ingredient composition, and nutritional databases, to optimize feed formulations. By considering factors such as animal age, weight, growth stage, and performance goals, AI can recommend personalized and precise nutrient compositions for individual animals or groups. This ensures that animals receive the necessary nutrients while minimizing waste and optimizing feed efficiency.
- **Feed quality assessment:** AI can help assess feed quality and safety by analyzing data from various sources, such as lab reports, ingredient specifications, and historical data. ML algorithms can identify patterns and anomalies that indicate variations in feed quality, such as contamination or nutrient deficiencies. AI-powered systems can provide real-time feedback on feed quality, enabling farmers to make informed decisions about feed purchases and adjust feed rations accordingly.
- **Real-time feed management:** AI can assist in real-time feed management by integrating data from sensors, feeding systems, and environmental conditions. By monitoring feed intake, animal behavior, and environmental factors, AI algorithms can optimize feed delivery schedules and

adjust feed quantities based on individual animal needs. This helps prevent overfeeding or underfeeding, reduces feed waste, and improves feed conversion efficiency.

- **Predictive modeling for feed optimization:** AI can develop predictive models that consider multiple variables, such as animal performance data, weather forecasts, feed prices, and ingredient availability. By analyzing historical and real-time data, AI algorithms can forecast animal growth, feed requirements, and production outcomes. This enables farmers to optimize feed management strategies, adjust feed rations, and make informed decisions on feed purchasing and allocation (Liebe and White, 2019).
- **Impact assessment:** AI can evaluate the environmental impact of feed management practices by analyzing data on feed ingredients, energy usage, water consumption, and greenhouse gas emissions. By assessing the lifecycle impacts of different feed formulations and management strategies, AI can help identify opportunities to minimize the environmental footprint of livestock production. This includes optimizing ingredient sourcing, reducing feed waste, and promoting sustainable feed production practices.

Automation for precision farming

Artificial intelligence—powered systems can automate various tasks in animal farming, such as feeding, milking, and waste management. Robotic devices equipped with AI capabilities can carry out these tasks efficiently and accurately, freeing up human labor (Carolan, 2020) and ensuring consistent and precise operations. AI can also integrate data from multiple sources, such as sensors and cameras, to provide real-time insights and automated decision-making. It can precision farming practices by optimizing resource allocation and utilization (Hamadani and Khan, 2015). By analyzing data on animal behavior, environmental conditions, and resource availability, AI algorithms can provide insights for efficient resource management. This includes optimizing grazing patterns, water usage, energy consumption, and waste management. Precision farming techniques can reduce environmental impact and promote sustainable livestock production.

AI can revolutionize automated monitoring and management systems in animal farming by enabling real-time data collection, analysis, and decision-making. Here is how AI can be applied in this context.

1. **Sensor-based data collection:** AI can leverage sensors and IoT devices to collect real-time data on various parameters, such as animal behavior, health indicators, environmental conditions, and feed consumption. Sensors can be integrated into wearable devices, feeding systems, environmental monitoring systems, and more. AI algorithms can process and analyze these data, providing valuable insights into animal well-being, productivity, and farm conditions.
2. **Behavior analysis:** AI algorithms can analyze sensor data to monitor animal behavior patterns and detect deviations from normal behavior. By identifying changes in activity levels, feeding patterns, or social interactions, AI can alert farmers to potential health issues, stress, or abnormal behavior that may indicate a problem. Timely intervention based on AI-driven behavior analysis can help prevent disease outbreaks, improve animal welfare, and optimize production outcomes.
3. **Automated systems:** AI can optimize and automate feeding systems in animal farming. By integrating data from sensors, animal profiles, and nutritional requirements, AI algorithms can adjust feed quantities, feeding schedules, and formulations based on individual animal needs and

performance goals. This ensures optimal feed utilization, reduces feed wastage, and promotes healthy growth. AI-powered feeding systems can also monitor feed availability and track consumption rates and robotic milking systems: Robotic milking systems are becoming more common in dairy farming (Hamadani and Khan, 2015). These systems use sensors and robotic arms to clean and milk cows without human intervention. They can monitor the cow's health and milk production and even track individual cow data.

4. Automated feeding systems are used in both dairy and livestock farming. These systems can automatically distribute feed to animals based on preprogrammed schedules or individual animal needs. They help optimize feed efficiency and reduce labor requirements. In poultry farming, automated systems are used for egg sorting and grading. These systems use computer vision and sensors to detect defects, measure size, and sort eggs based on various parameters such as weight and quality. AI can optimize environmental conditions in animal farming by monitoring and controlling factors such as temperature, humidity, ventilation, and lighting. By analyzing real-time sensor data and historical patterns, AI algorithms can adjust environmental parameters to provide optimal conditions for animal comfort, health, and productivity. Automated systems controlled by AI can regulate environmental parameters, detect abnormalities, and notify farmers in case of deviations that may impact animal well-being.
5. **Animal farm robots:** Robots are being increasingly used on animal farms to streamline operations, enhance productivity, and improve animal welfare. A few robots developed for animal farming are listed in Table 11.1.

Genetic improvement and breeding

Animal genetics and breeding is a data-intensive science and techniques like the least squares, and best linear unbiased predictions have been used over many years for selection and breeding (Hamadani et al., 2019, 2020a, 2021a; Rather et al., 2019; Baba et al., 2020a,b). AI can accelerate genetic improvement in animal breeding programs by analyzing large genomic datasets. ML algorithms can identify genetic markers associated with desired traits, predict breeding values, and optimize mating strategies. This enables breeders to make informed decisions to enhance desirable traits, reduce genetic disorders, and improve overall breeding efficiency (Hamadani et al., 2022b,c). Implementing AI in genetic improvement and breeding programs requires reliable genomic data, advanced AI algorithms, and collaborations between breeders, researchers, and technology providers. It is crucial to ensure data privacy, maintain genetic diversity, and consider ethical and welfare aspects associated with breeding practices. By leveraging AI in genetic improvement and breeding, animal farmers can accelerate genetic progress, enhance production traits, and achieve sustainable and resilient livestock populations.

1. **Genomic selection:** AI algorithms can analyze large-scale genomic data to predict the genetic potential of animals. By leveraging ML techniques, AI can identify genetic markers associated with desirable traits, such as milk production, meat quality, disease resistance, or reproductive performance. AI-powered genomic selection models can assist breeders in selecting animals with the highest genetic merit for breeding, accelerating the rate of genetic improvement in livestock populations.

Table 11.1 Some robots used in animal farms. Genetic improvement and breeding.

Robot name	Utility	References
MilkoMatic	MilkoMatic is a state-of-the-art milking robot designed to automate the milking process on dairy farms. Equipped with advanced sensors and robotic arms, MilkoMatic can identify cows, attach to their udders, and efficiently milk them. It also monitors milk quality, ensuring optimal hygiene standards.	https://milkomatic.com/
Sheepdog robot	Sheepdog robot is a vision-based robot for detecting different predators among livestock in pastures and also distinguishes them from other animal species.	Riego del Castillo et al. (2022)
FeedMaster	FeedMaster is a feeding robot that ensures precise and timely distribution of feed on animal farms. It employs advanced programming and sensors to accurately dispense the right amount of feed to each animal. FeedMaster also monitors consumption patterns, enabling personalized nutrition plans for improved animal health.	Kabir et al. (2022)
Clean Sweep	Clean Sweep is a robotic cleaner designed to maintain cleanliness in barns and pens. This agile robot autonomously navigates the area, collecting animal waste and debris. Equipped with efficient waste disposal mechanisms, clean Sweep promotes a hygienic environment, minimizing the risk of diseases.	Shukla et al. (2022)
T-Moov	T-Moov T- Moov, created by Tibot, is a revolutionary poultry robot designed to address the specific needs of poultry farmers. It has been meticulously developed in collaboration with industry experts to excel in challenging environments characterized by factors like poor litter conditions or flat feed chains. The primary objective of T- Moov is to ensure continuous movement among your birds and minimize the occurrence of floor eggs.	https://www.tibot.fr/en/solutions/poultry-robot-tmoov/
Xo robot	Developed by Octopus poultry robotics. XO is an invaluable tool for poultry farmers, providing assistance in risk prevention and contamination treatment. It ensures continuous treatment of litter and diligent monitoring of the environment and the well-being of the animals.	https://www.octopusbiosafety.com/es/salud-animal/xo/

2. **Breeding value estimation:** AI can estimate breeding values, which reflect an animal's genetic superiority for specific traits, by integrating data from pedigree records, phenotypic data, and genomic information. AI algorithms can consider complex genetic interactions and environmental factors to calculate accurate breeding values. This enables breeders to make informed decisions regarding mating pairs, selection of breeding stock, and genetic progress tracking.
3. **Genetic diversity analysis:** AI algorithms can analyze genomic data to assess the genetic diversity within a population and identify inbreeding levels. By evaluating patterns of genetic variation and relatedness, AI can help breeders maintain optimal levels of genetic diversity to prevent inbreeding depression and promote the long-term health and performance of the population. AI-powered systems can provide recommendations on mating strategies and breeding plans to manage genetic diversity effectively.

4. **Parentage verification and pedigree reconstruction:** AI can utilize genomic data to verify parentage and reconstruct pedigrees accurately. By comparing genetic profiles, AI algorithms can determine parent–offspring relationships, identify potential errors in pedigree records, and assist in pedigree reconstruction. This ensures accurate genetic evaluations and supports the selection of superior breeding stock.
5. **Trait prediction and optimization:** AI can predict the performance of animals for various traits based on genetic and environmental data. By considering factors such as genotype, management practices, and environmental conditions, AI algorithms can provide insights into the potential performance of offspring for specific traits. This helps breeders optimize mating decisions, select animals with desired trait combinations, and design breeding programs targeted toward specific production goals.

Decision support systems

Artificial intelligence–based decision support systems can provide real-time recommendations and predictions to farmers. By integrating data on animal health, environmental conditions, and market trends, these systems can assist in making informed decisions related to breeding, nutrition, disease management, and resource allocation. This helps optimize farm operations, improve productivity, and enhance profitability (Hamadani and Ganai, 2022). AI-powered decision support systems can greatly enhance decision-making processes in animal farming. Here is how AI can be applied in decision support systems for animal farm management.

1. **Data integration:** AI can integrate and analyze data from various sources, including sensor data, farm management systems, weather forecasts, market trends, and genetic information. By aggregating and processing these data, AI algorithms can provide a comprehensive view of the farm's operations and performance. This enables farmers to make informed decisions based on a holistic understanding of multiple factors influencing their operations.
2. **Performance monitoring and analysis:** AI can monitor and analyze key performance indicators (KPIs) related to animal health, growth, production, and efficiency. By processing real-time and historical data, AI algorithms can detect trends, identify areas of improvement, and provide insights into farm performance. Farmers can track KPIs, such as feed conversion ratio, growth rates, disease incidence, and reproductive efficiency, to optimize management strategies and improve overall farm productivity.
3. **Risk assessment and management:** AI can assess and manage risks associated with animal farming. By analyzing data on disease prevalence, weather conditions, market prices, and financial indicators, AI algorithms can identify potential risks and provide risk mitigation strategies. AI-powered systems can assist in making decisions related to disease prevention, biosecurity measures, market timing, and financial planning, reducing uncertainties and improving farm resilience.
4. **Resource optimization:** AI can optimize resource allocation and utilization in animal farming. By analyzing data on feed availability, energy consumption, labor costs, and equipment usage, AI algorithms can provide recommendations for efficient resource management. Farmers can optimize feed rations, adjust staffing levels, schedule maintenance tasks, and plan production

cycles based on AI-driven insights, leading to improved cost-effectiveness and resource efficiency.

5. **Financial analysis and forecasting:** AI can assist in financial analysis and forecasting for animal farming operations. By integrating data on costs, revenues, market trends, and production outputs, AI algorithms can generate financial models and forecasts. This helps farmers evaluate profitability, assess the economic impact of management decisions, and plan for future investments or expansions.
6. **Real-time recommendations and alerts:** AI-powered decision support systems can provide real-time recommendations and alerts to farmers. By analyzing real-time sensor data, market information, and performance indicators, AI algorithms can suggest immediate actions or adjustments. For example, AI systems can provide alerts for deviations in environmental parameters, recommend adjustments to feed quantities, or suggest timely interventions for health issues. Real-time recommendations enable proactive decision-making and rapid responses to optimize farm performance.

Improving animal production using AI

Artificial intelligence can play a significant role in the production and optimization of animal products in several ways (Aharwal et al., 2021), the most important being the production of animal protein. This is done by optimizing the four most important aspects of animal farming viz feeding, breeding, weeding, and heeding. This is achieved through the following management practices:

1. **Quality control and traceability:** AI technologies can improve quality control and traceability in animal production. Computer vision systems can be used to identify and sort meat products based on quality parameters such as color, marbling, and texture. AI algorithms can also track and trace animal products throughout the supply chain, providing information on the origin, production methods, and safety standards. This helps ensure product integrity, reduce fraud, and enhance consumer confidence.
2. **Sustainability and resource optimization:** AI can contribute to the sustainable production of animal protein by optimizing resource utilization and reducing environmental impact. AI-powered systems can monitor and optimize energy usage, water consumption, and waste management in animal farming operations. By analyzing data on environmental conditions, feed efficiency, and production parameters, AI algorithms can identify opportunities to reduce resource waste, minimize greenhouse gas emissions, and improve overall sustainability (Nti et al., 2022).
3. **Processing and manufacturing:** AI technologies can automate and optimize various processes involved in the production of animal products. Computer vision systems can be used to assess the quality of raw materials, such as meat or dairy, and guide automated sorting and processing procedures. AI algorithms can analyze data from sensors and imaging technologies to monitor production parameters, such as temperature, humidity, and processing time, ensuring consistent quality and reducing product defects.
4. **Quality control and inspection:** AI can improve quality control processes by analyzing data from visual inspections, sensors, and other measurement devices. Computer vision algorithms can identify and classify defects, such as blemishes or abnormalities in animal products, allowing

for real-time quality assessments and automatic rejection of substandard items. This helps maintain consistent quality standards and reduces the risk of defective products reaching consumers.

5. **Food safety and traceability:** AI technologies can enhance food safety measures in animal products. ML algorithms can analyze data from various sources, including production records, supply chain information, and historical data on safety incidents, to identify patterns and potential risks. AI-powered systems can help detect and prevent foodborne illnesses, ensure compliance with safety regulations, and provide traceability information to track products from farm to fork.
6. **Sensory evaluation and consumer experience:** AI can contribute to sensory evaluation and product development for animal products. ML algorithms can analyze sensory data, such as taste, texture, and aroma, to identify consumer preferences and develop optimized product formulations. AI can also assist in personalized recommendations, such as suggesting suitable meat cuts or dairy products based on individual preferences and dietary requirements.
7. **Product labeling and allergen detection:** AI can aid in accurate labeling and allergen detection in animal products. Natural language processing algorithms can analyze product labels and ingredient lists, ensuring compliance with labeling regulations and identifying potential allergens. This helps prevent mislabeling and supports consumers with dietary restrictions or allergies in making informed purchasing decisions.
8. **Consumer insights and market trends:** AI technologies can analyze vast amounts of data from various sources, such as social media, online reviews, and consumer feedback, to gain insights into consumer preferences and market trends. This information can help animal product manufacturers understand consumer demands, develop targeted marketing strategies, and tailor their products to meet evolving consumer needs.

By leveraging AI in the production and management of animal products, businesses can enhance product quality, safety, and consumer satisfaction. However, it is important to ensure data privacy, comply with regulatory requirements, and address ethical considerations associated with AI implementation in the food industry.

Conclusion

The integration of AI into animal management and breeding has emerged as a transformative approach with immense potential. AI offers the capability to analyze vast amounts of data from various sources, including genomic information, sensor data, and historical records, to drive informed decision-making and improve animal productivity, health, and welfare. By leveraging advanced algorithms and ML techniques, AI can enhance precision in breeding programs by identifying desirable genetic traits, predicting breeding values, and optimizing mating strategies. AI-powered systems enable real-time monitoring of animal health, behavior, and environmental conditions, facilitating early disease detection, targeted interventions, and optimized resource allocation. Additionally, AI-driven decision support systems aid in risk assessment, financial analysis, and resource optimization, thereby promoting sustainability, efficiency, and profitability in animal farming. With its ability to handle complex datasets and generate actionable insights, AI is poised to revolutionize animal management and breeding practices, driving advancements in livestock production, genetic improvement, and sustainable farming practices.

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